

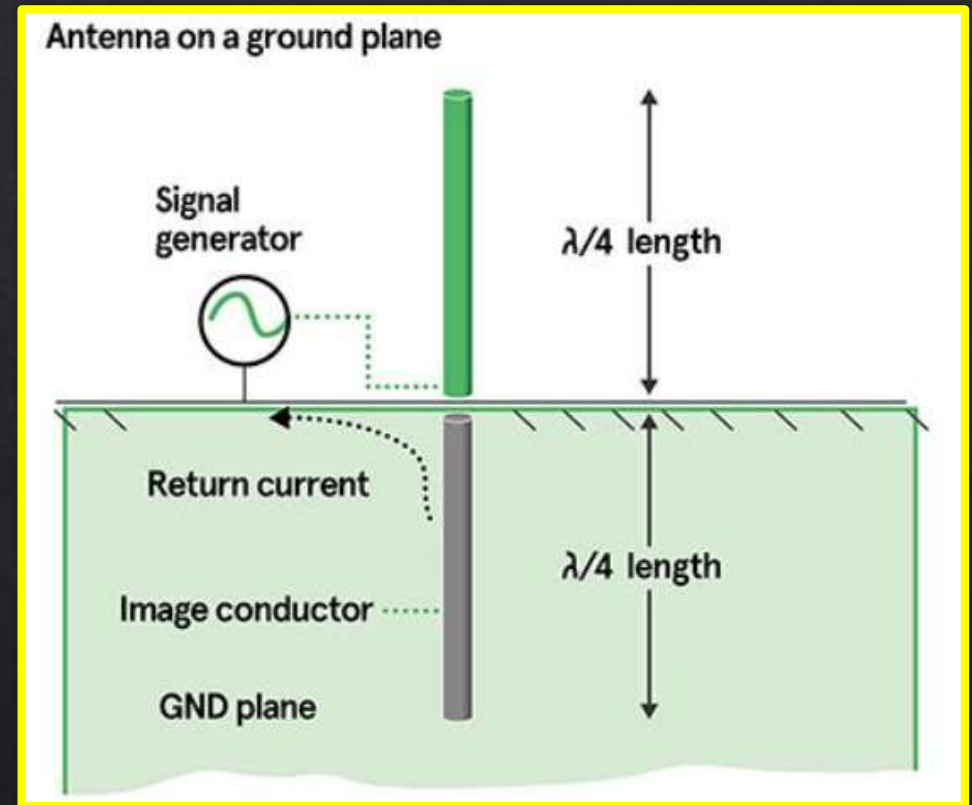
UWB Monopole antennas

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What is a monopole antenna ?

A monopole antenna is a simple type of antenna that consists of a single conducting element, typically a vertical metal strip or wire, placed above a ground plane. It can be understood as half of a dipole antenna, where the ground plane acts as a mirror that creates an image of the missing half. When the antenna is excited, current flows upward along the radiator and returns through the ground plane, forming a complete current path.

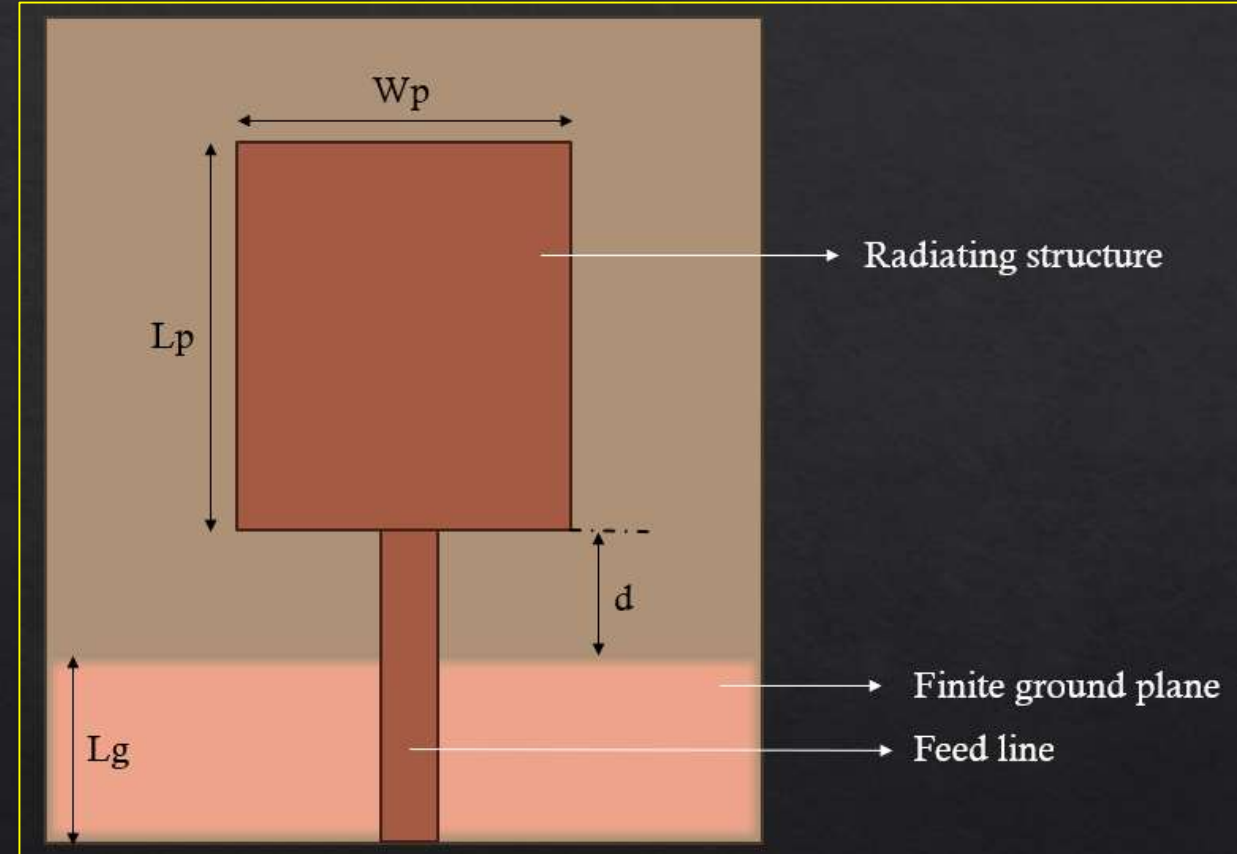
This flow of current produces electromagnetic radiation. Because the current is mainly vertical, the antenna radiates energy uniformly in all horizontal directions, resulting in an omnidirectional radiation pattern. The size and shape of the ground plane play an important role, as they influence the current distribution and overall radiation performance of the antenna.



Key factors for a monopole antenna

1. Patch length (L_p):

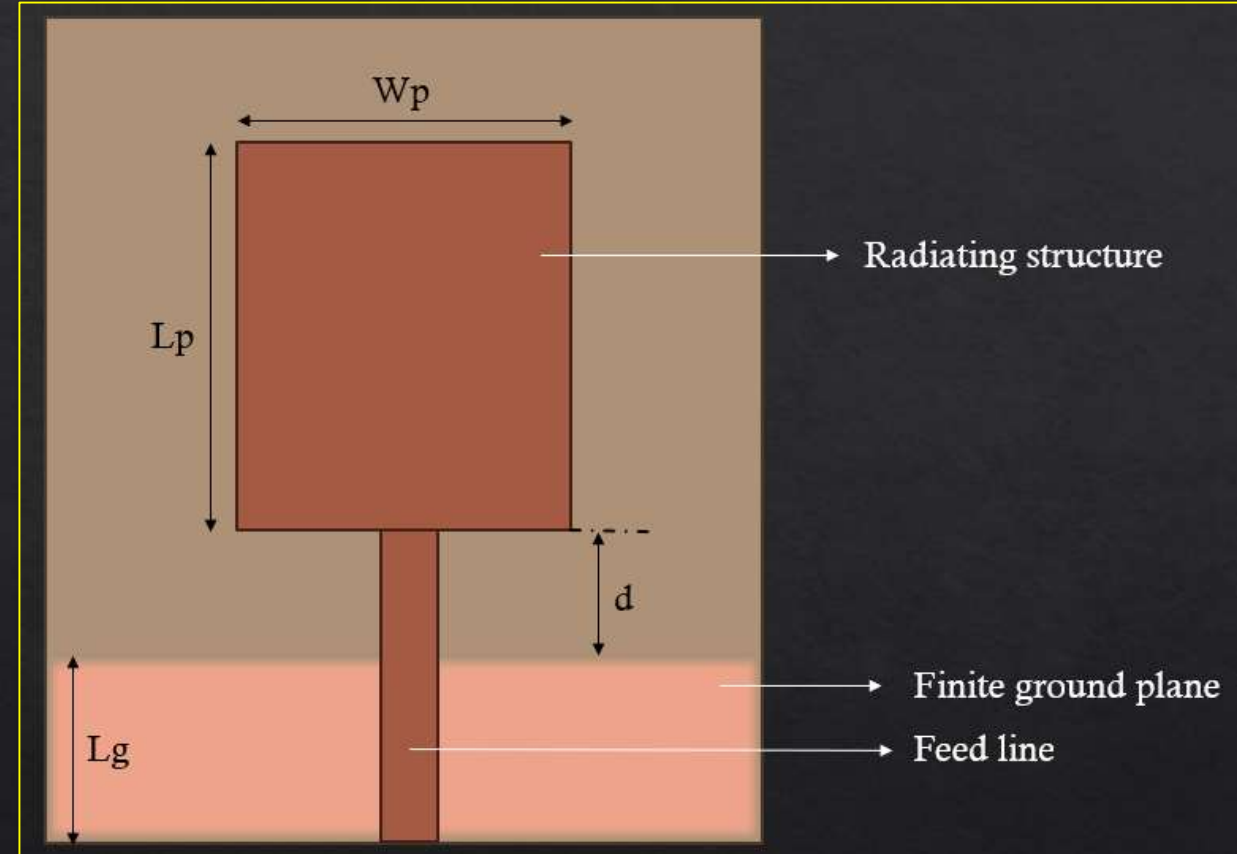
The patch length is the primary parameter that determines the resonant frequency of the antenna. It defines the effective current path along the radiator, and for a monopole antenna, this length is typically close to a quarter of the operating wavelength. When the patch length is increased, the current travels a longer path, which lowers the resonant frequency.



Key factors for a monopole antenna

2. Gap between ground and radiator edge (d):

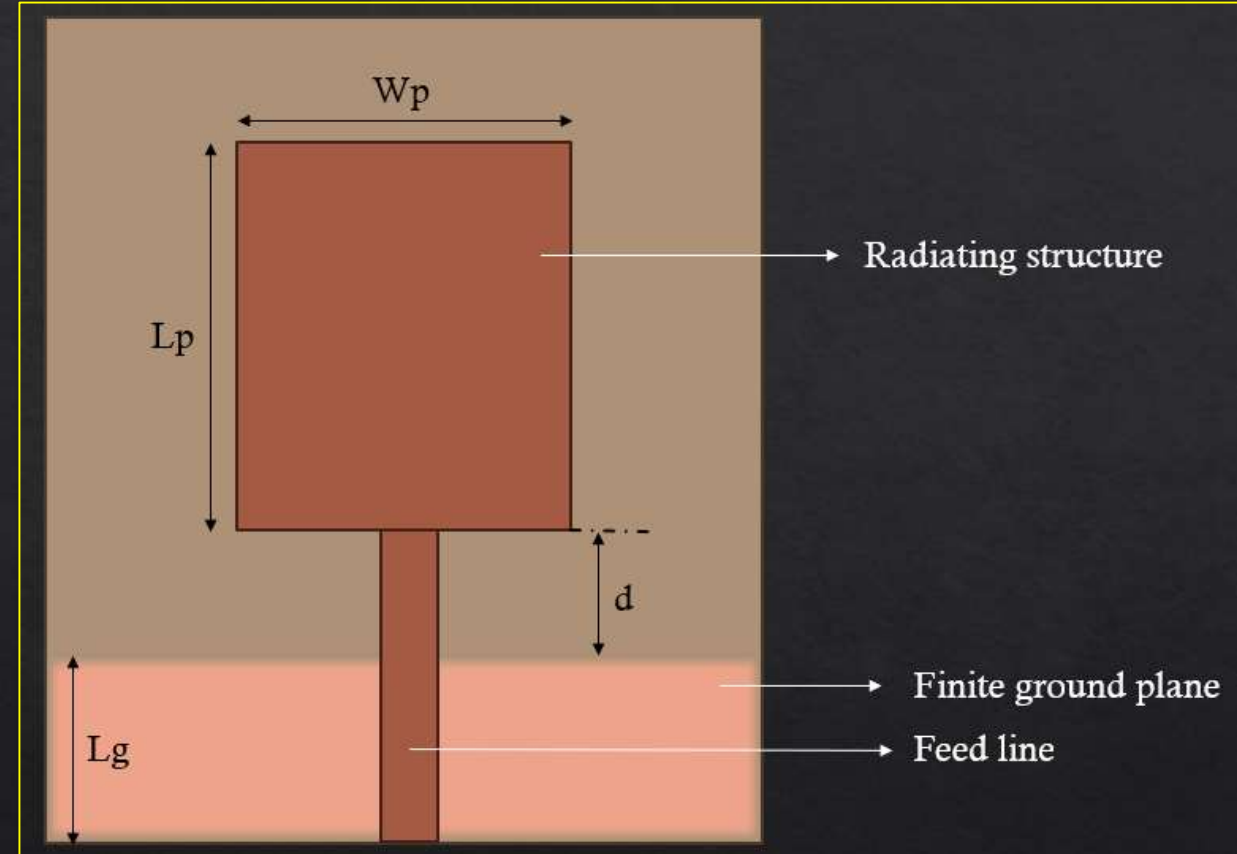
The gap between the lower edge of the patch and the upper edge of the ground plane acts as a critical coupling region. This small spacing introduces a capacitive effect that influences how efficiently power is transferred from the feed to the radiator. If the gap is too small, the coupling becomes too strong, which can lead to impedance mismatch and narrow bandwidth. If the gap is too large, the coupling weakens, and the antenna may not be properly excited. An optimized gap ensures proper impedance matching and is essential for achieving wideband or ultra-wideband performance.



Key factors for a monopole antenna

3. Length of ground (L_g):

The ground plane in a printed monopole is intentionally truncated and plays an active role in radiation. It serves as the return path for current and also contributes to the overall radiation mechanism. If the ground plane is too large, the antenna begins to behave like a microstrip patch antenna, resulting in narrowband performance. If it is too small, the return current becomes insufficient, reducing radiation efficiency. An appropriately chosen ground plane length allows proper current distribution and helps in generating multiple resonant modes, which is important for broadband operation.



Reference Paper

The antenna presented in the paper is a **microstrip patch antenna**, not a printed monopole, as it employs a full ground plane and operates based on cavity-type resonant modes rather than monopole radiation. The design uses a **Ψ -shaped (psi-shaped) patch geometry** to achieve broadband performance in the microwave range, with operating frequencies approximately spanning **4–7 GHz**. The unique Ψ structure introduces multiple current paths on the patch, which in turn excites several resonant modes at different frequencies. The central arm and side branches of the Ψ shape create distinct current distributions, leading to resonances around ~ 4.3 GHz, ~ 5.5 GHz, and ~ 7 GHz.

These modes are closely spaced and overlap, resulting in an overall wide impedance bandwidth. The antenna is typically probe-fed, and a tuning stub is used to improve impedance matching. The broadband behavior is therefore achieved not through monopole action, but through **multi-mode excitation within the patch structure**, enabled by the geometry of the Ψ -shaped radiator.

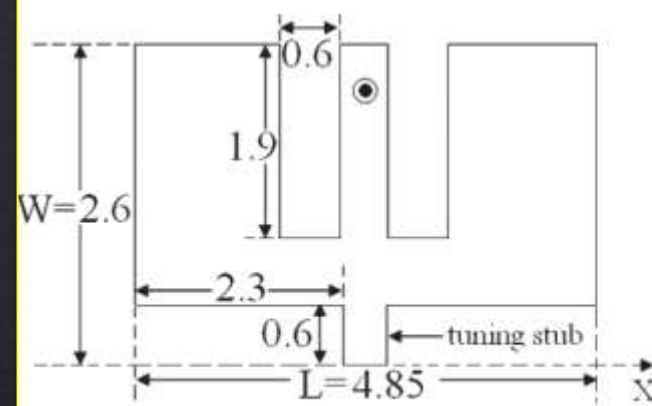


Figure 1a. A top view of the Ψ -shaped microstrip antenna [19].

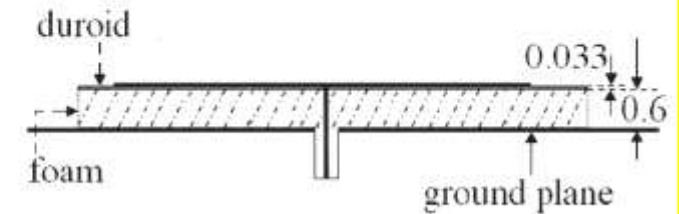


Figure 1b. A side view of the Ψ -shaped microstrip antenna [19].

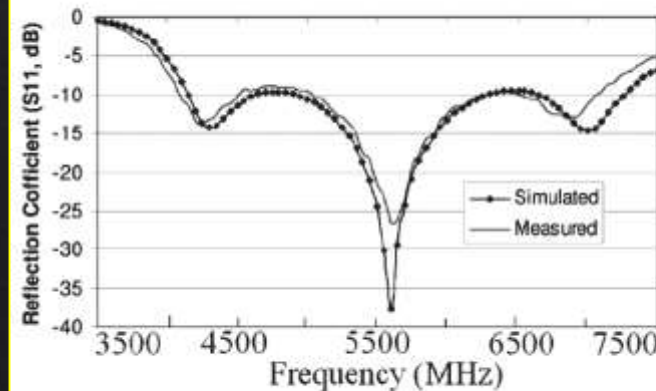


Figure 1c. The return loss over the bandwidth for the Ψ -shaped microstrip antenna [19].

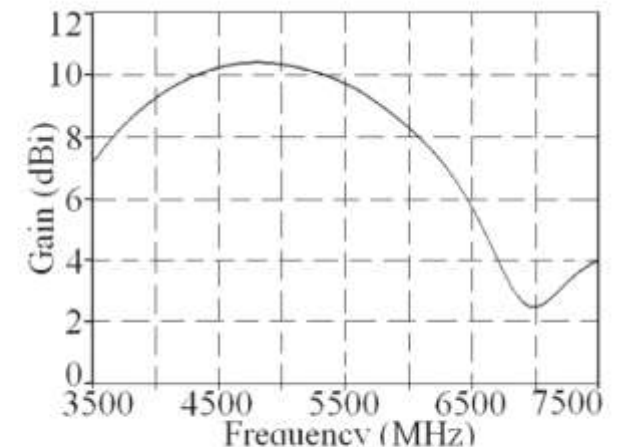
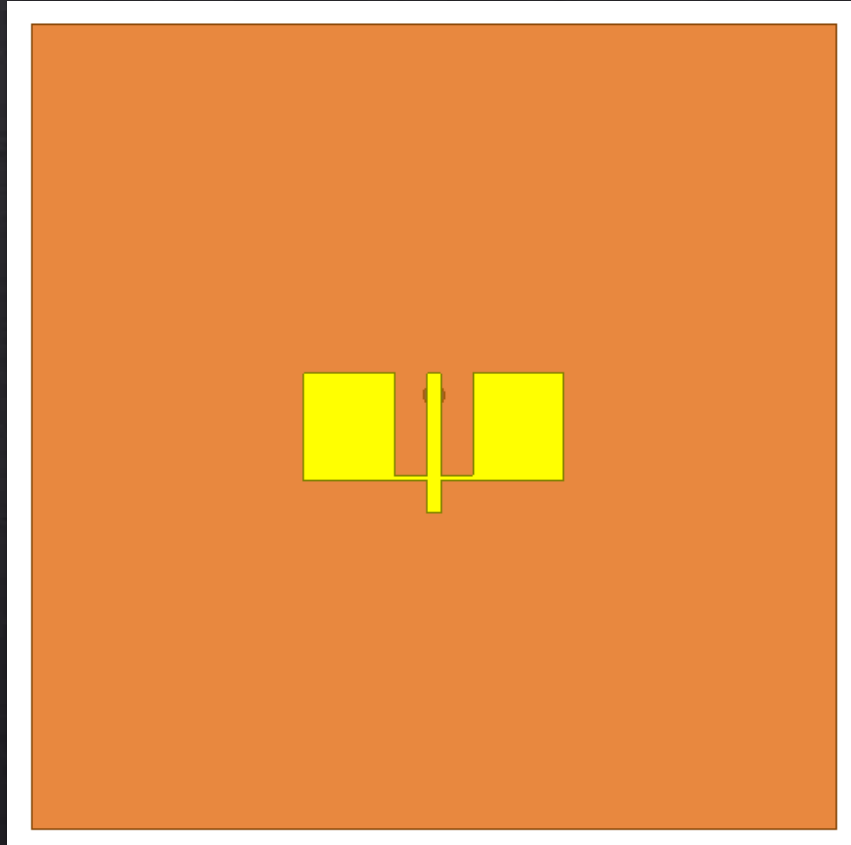
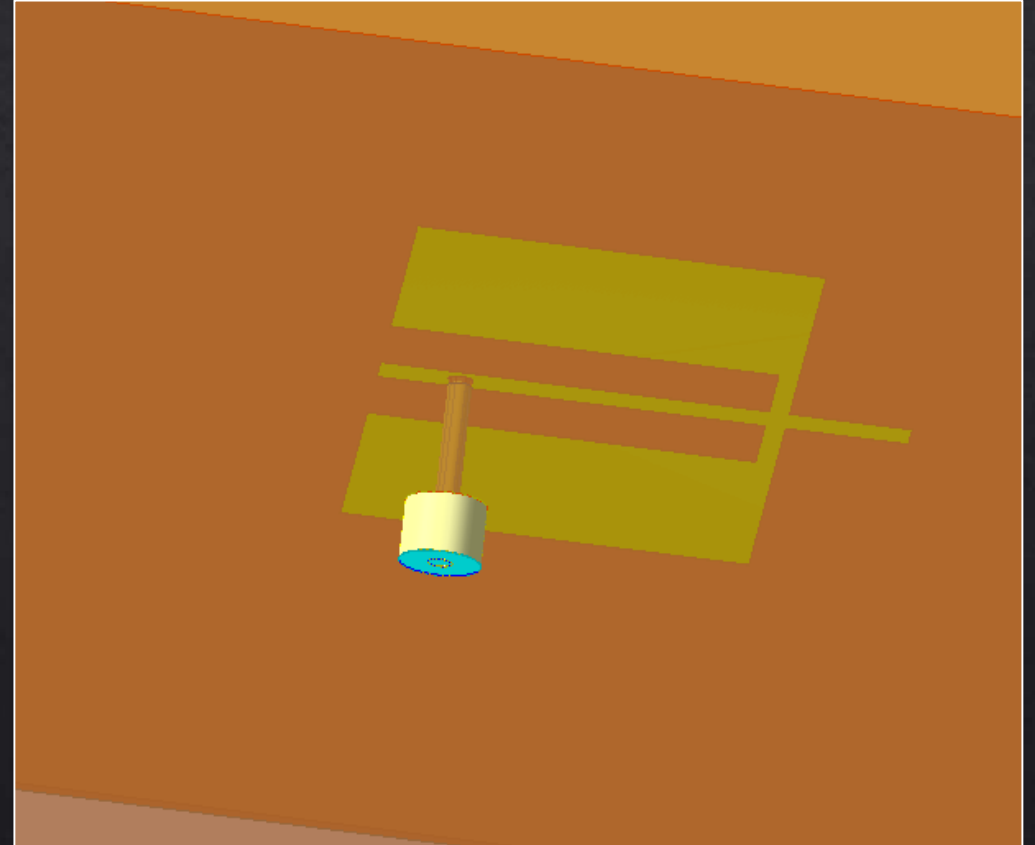


Figure 1d. The gain variation over the bandwidth for the Ψ -shaped microstrip antenna [19].

Simulation of the reference paper design

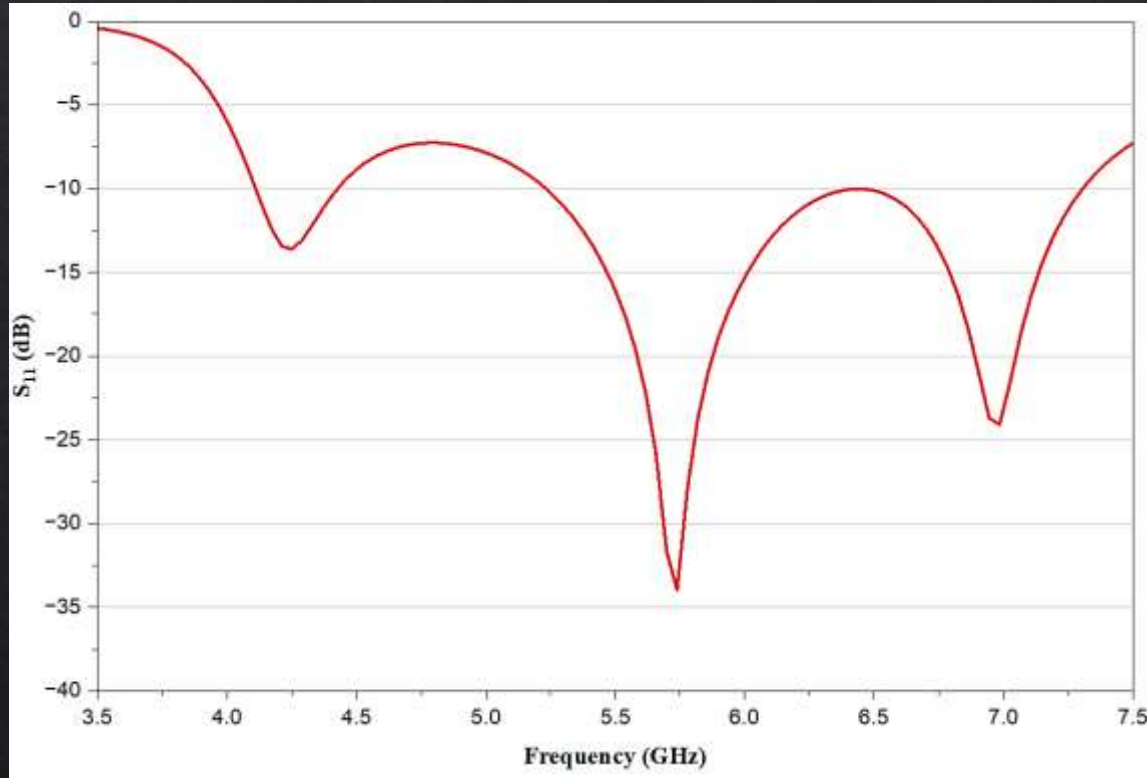


Top view of the patch

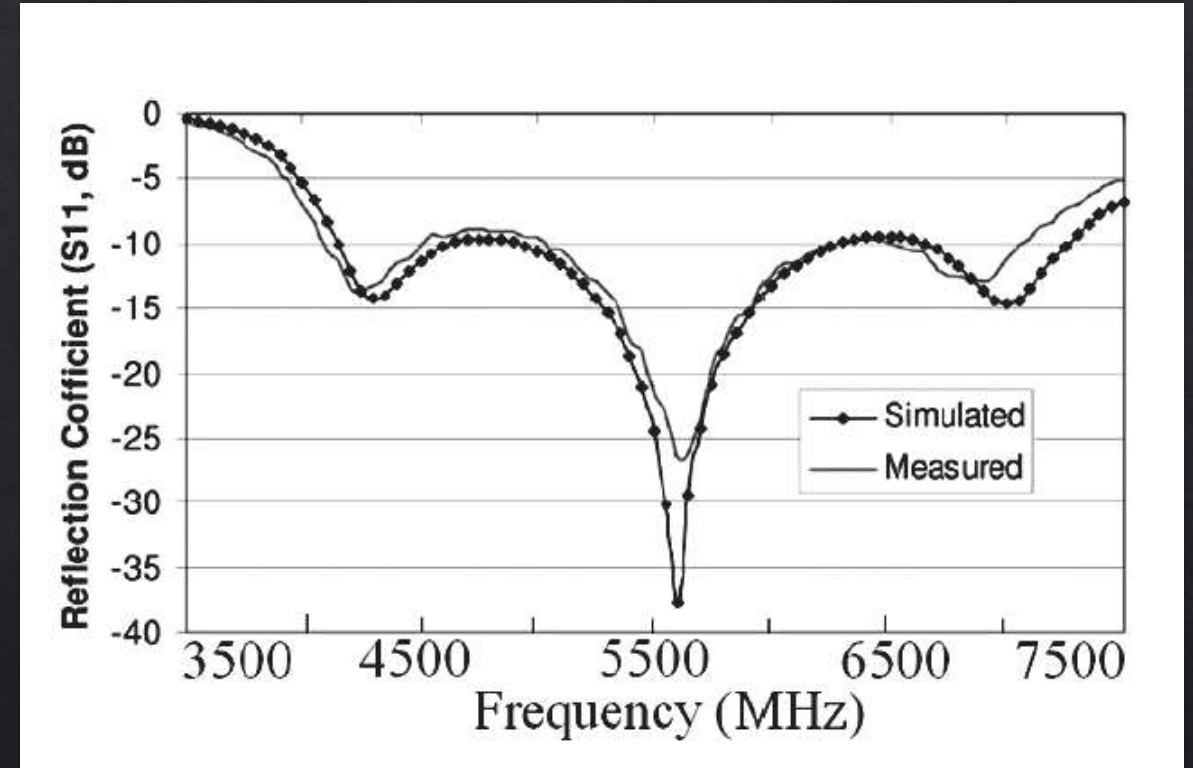


Bottom view showing the coax feed

Simulation of the reference paper design

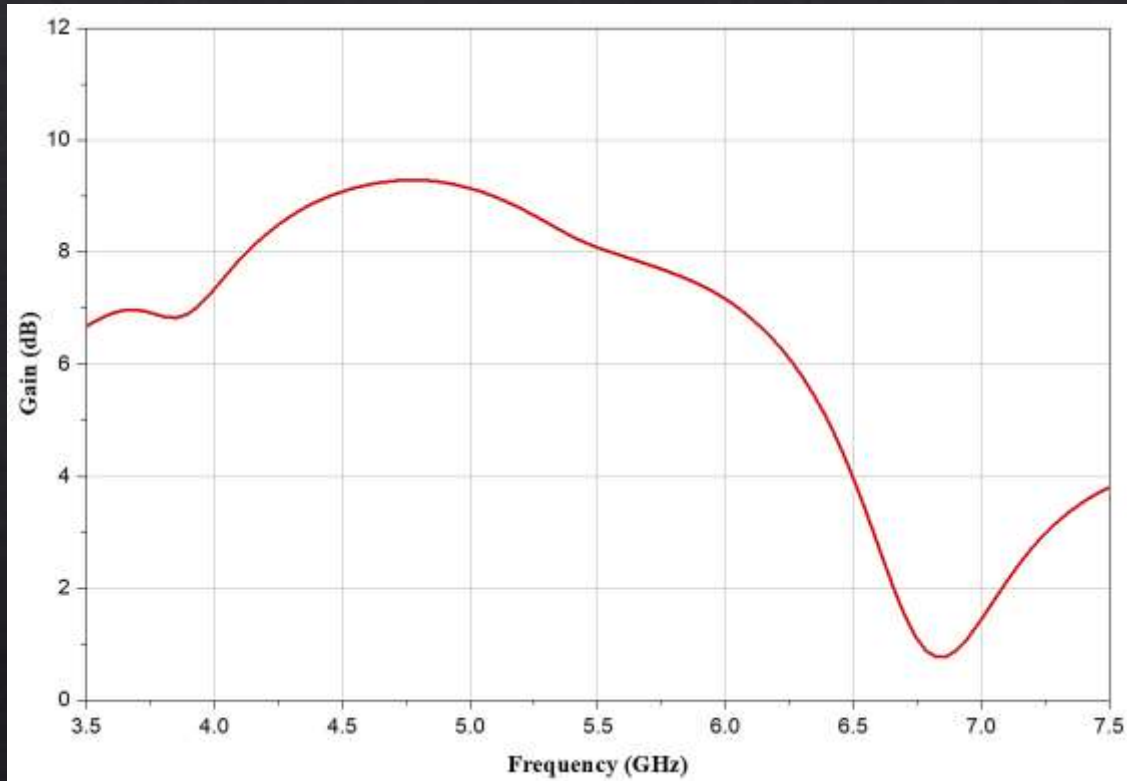


The S_{11} for the Ψ -shaped microstrip antenna over the bandwidth as per simulation.

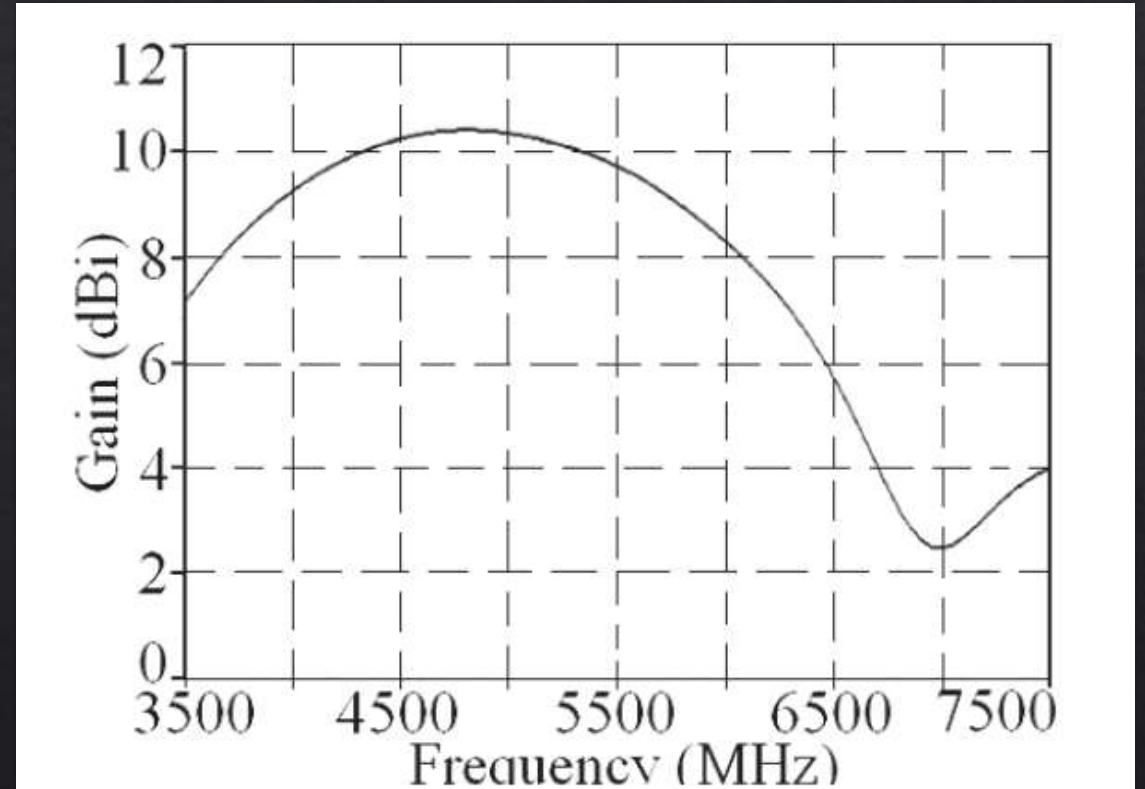


The S_{11} for the Ψ -shaped microstrip antenna over the bandwidth as per reference.

Simulation of the reference paper design

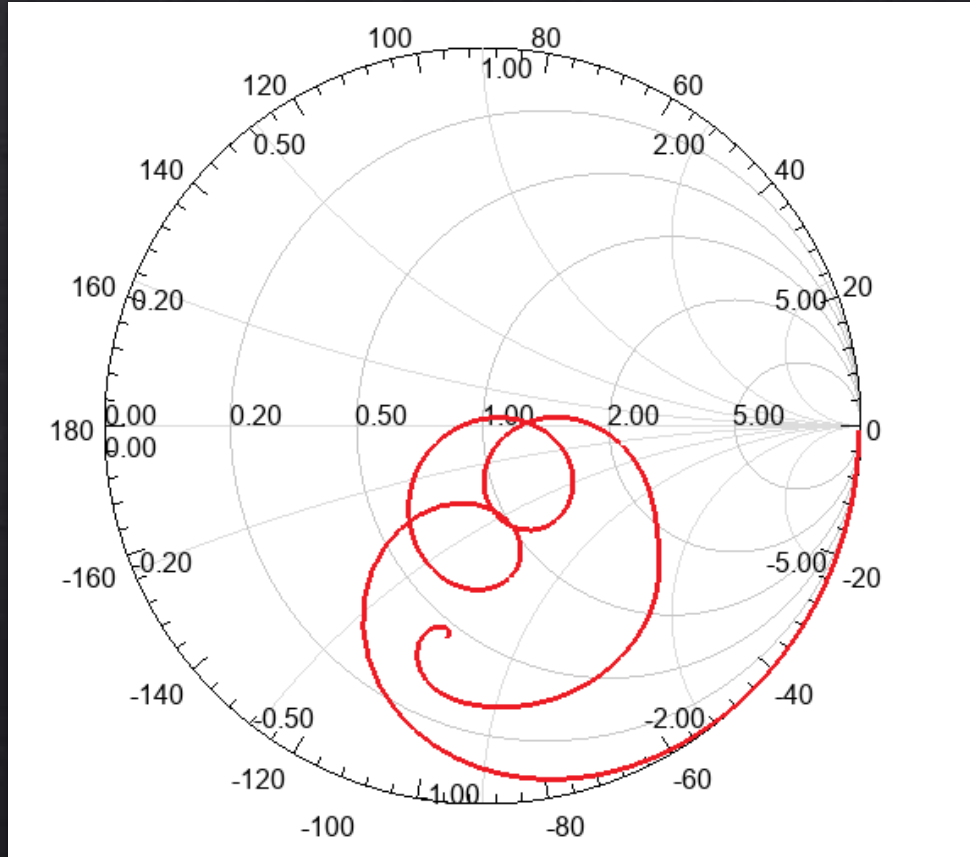


The simulated total broadside gain for the Ψ -shaped microstrip antenna over the bandwidth.

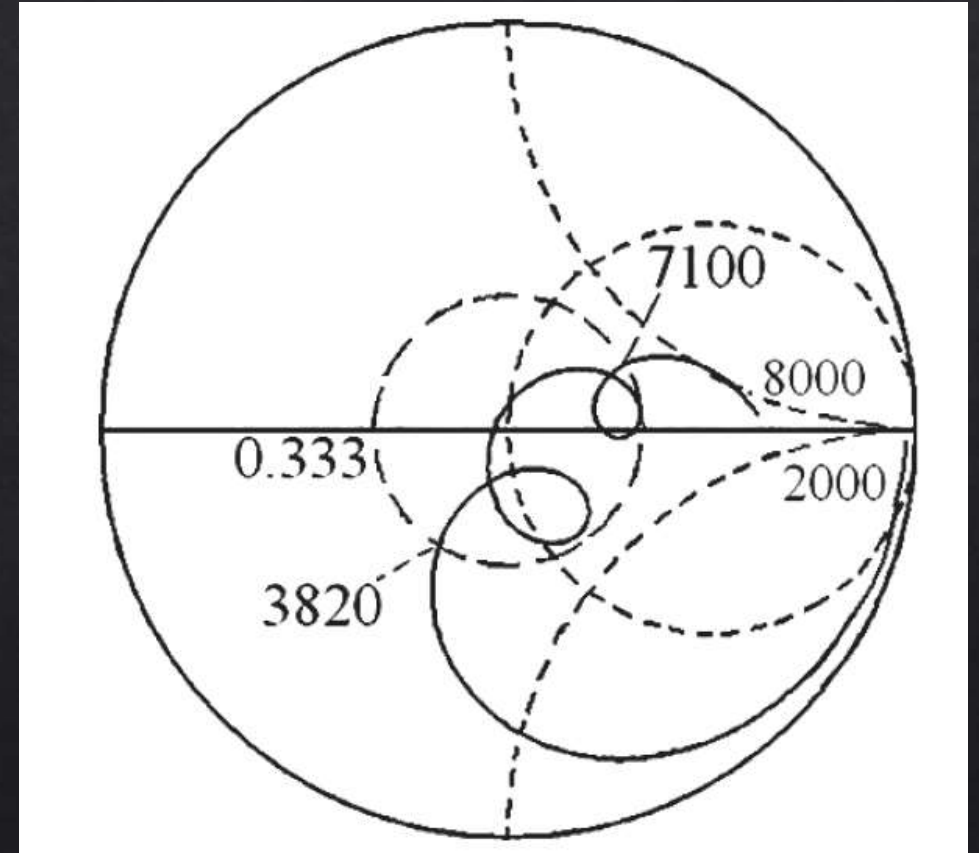


The total broadside gain for the Ψ -shaped microstrip antenna over the bandwidth as per reference.

Simulation of the reference paper design



The simulated input impedance for the Ψ -shaped microstrip antenna over the bandwidth.

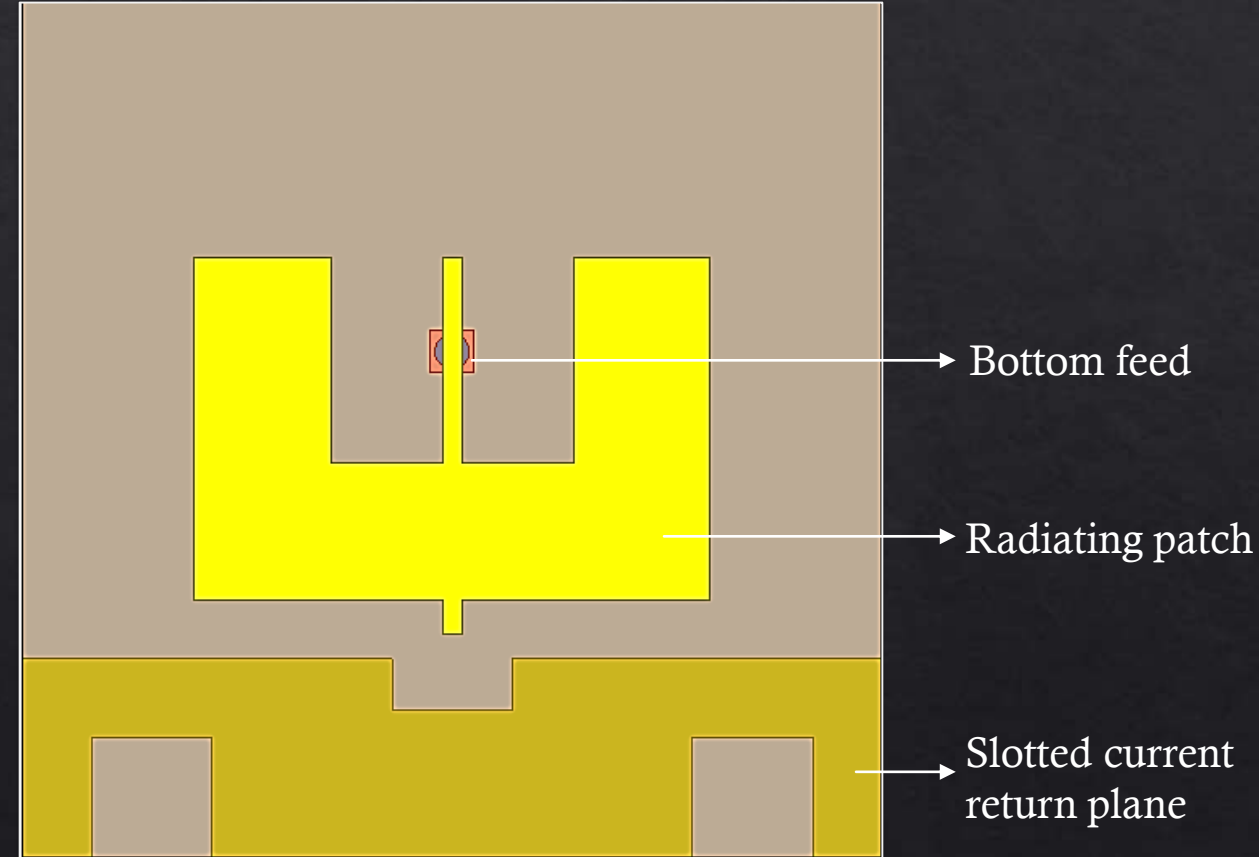


The input impedance for the Ψ -shaped microstrip antenna over the bandwidth as per reference.

Remodeling the reference model as a monopole

To remodel the given Ψ -shaped microstrip patch antenna as a **bottom-fed planar monopole** at 5.5 GHz, the full ground plane is removed and replaced with a **small, localized ground pad around the probe feed**, making it electrically small so that cavity modes are suppressed. The probe is retained to feed the radiator from below, with the outer conductor connected to this ground pad. The Ψ -shaped patch now acts as the monopole radiator, and its **effective length should be maintained close to $\lambda/4$** at 5.5 GHz to excite the fundamental monopole mode.

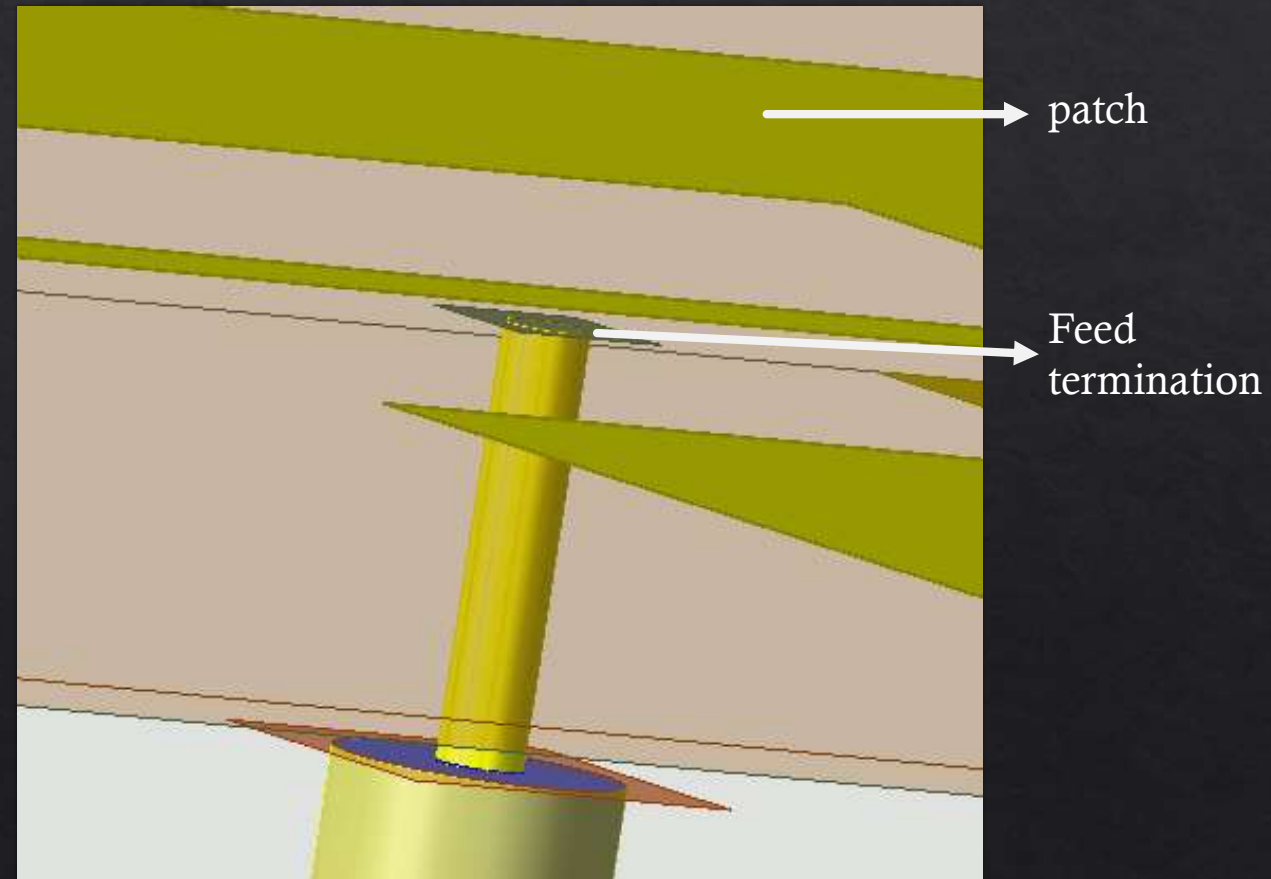
A **small gap ($\approx 1\text{--}3\text{ mm}$)** is introduced between the lower edge of the radiator and the ground pad to control capacitive coupling and impedance matching. The **ground length is kept minimal ($\ll \lambda/4$)** to avoid patch-like behavior, while the radiator dimensions are slightly tuned to maintain resonance. Proper optimization of the **patch length, feed gap, and ground size** ensures that the antenna transitions from a cavity-based patch to a radiating monopole with similar operating frequency.



Modelling of the feed line

Proximity feed

A proximity-fed microstrip patch antenna operates using capacitive coupling between the feed and the patch, rather than a direct electrical connection. This can be modeled using an equivalent circuit where the patch behaves as a parallel RLC resonator, and the feed is coupled through an additional capacitor representing the gap between them. This coupling capacitor controls the flow of energy into the patch, preventing abrupt excitation and reducing the amount of energy stored in the resonant cavity. As a result, the quality factor (Q) of the antenna decreases, leading to a less sharp resonance and a smoother impedance variation with frequency. This reduction in Q enables the antenna to maintain good impedance matching over a wider frequency range, thereby improving the -10 dB bandwidth compared to a directly fed patch antenna.



Modelling of the ground plane

Symmetric slotted ground

The ground plane of the proposed monopole antenna has been intentionally modified by introducing **rectangular slots**, which play a crucial role in improving the radiation characteristics. These slots disrupt the otherwise continuous current flow on the ground plane and help **reduce intersecting and circulating current paths**, which are a primary source of cross-polarized radiation. By guiding the return currents along more controlled and symmetric paths, the ground structure minimizes unwanted transverse current components that contribute to cross-polarization. As a result, the antenna exhibits improved polarization purity, with a noticeable reduction in the cross-polarized fields in the radiation pattern. Additionally, the slots also aid in modifying the effective current distribution on the ground, which can contribute to better impedance matching and bandwidth enhancement. Overall, the defected ground structure helps achieve a more stable and cleaner radiation behavior.



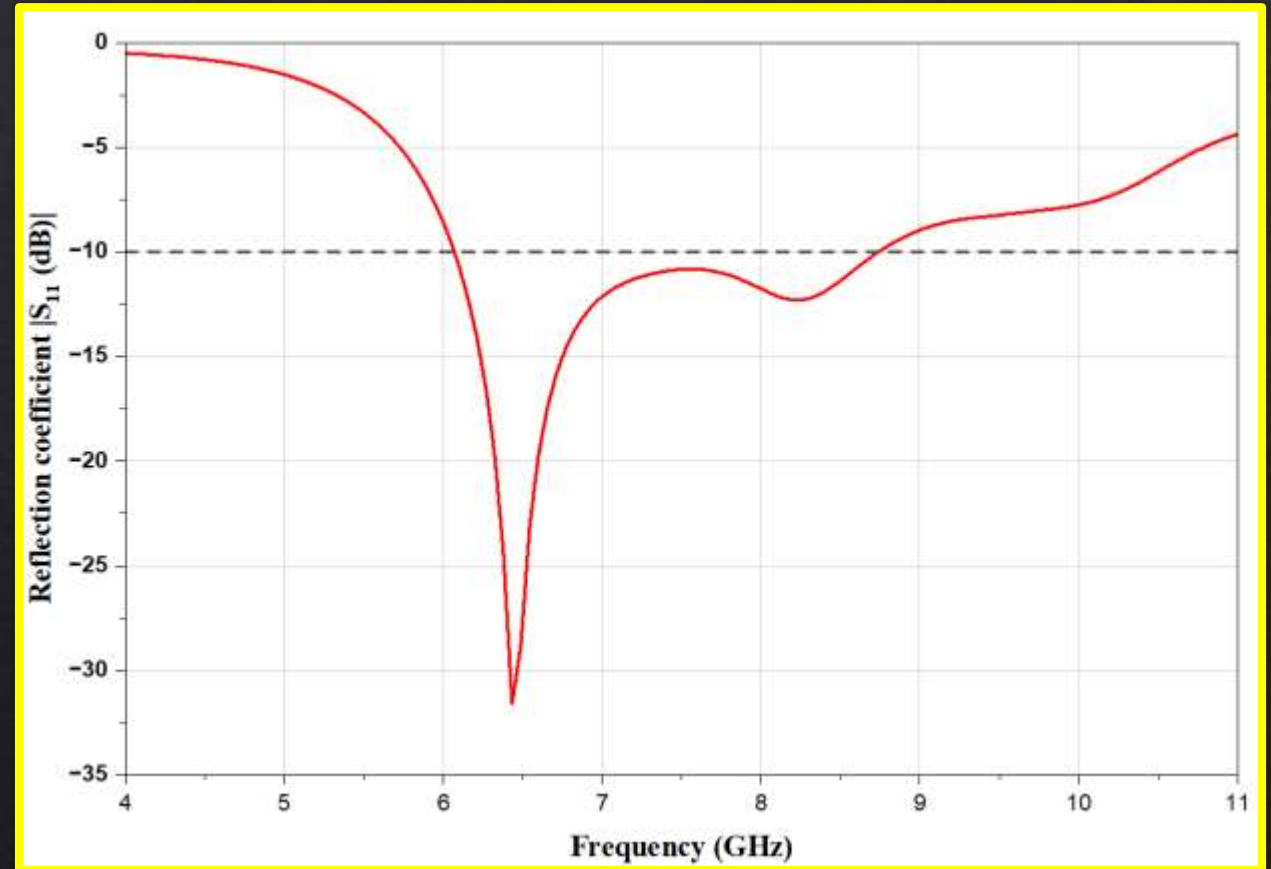
Current return plane

slots

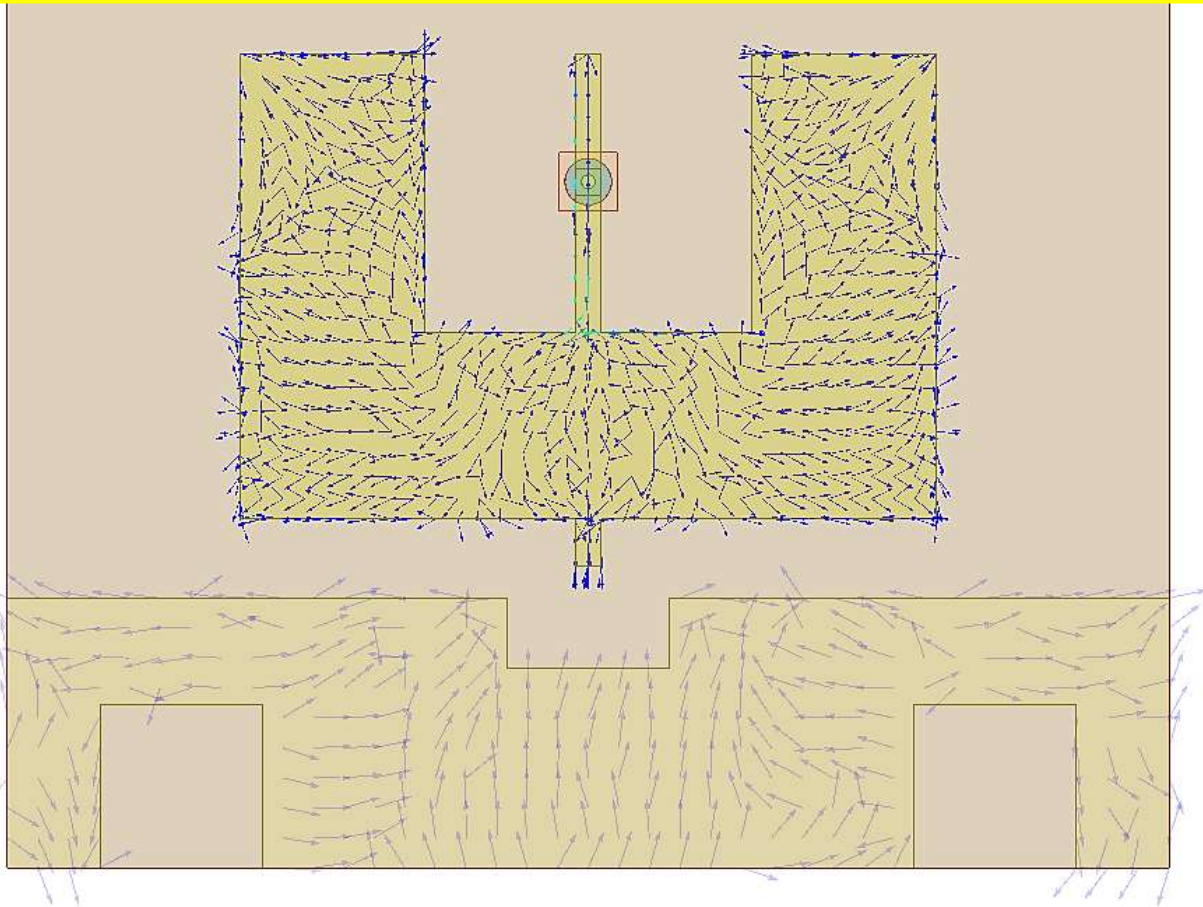
Simulation results

Reflection coefficient [S_{11} (dB)]

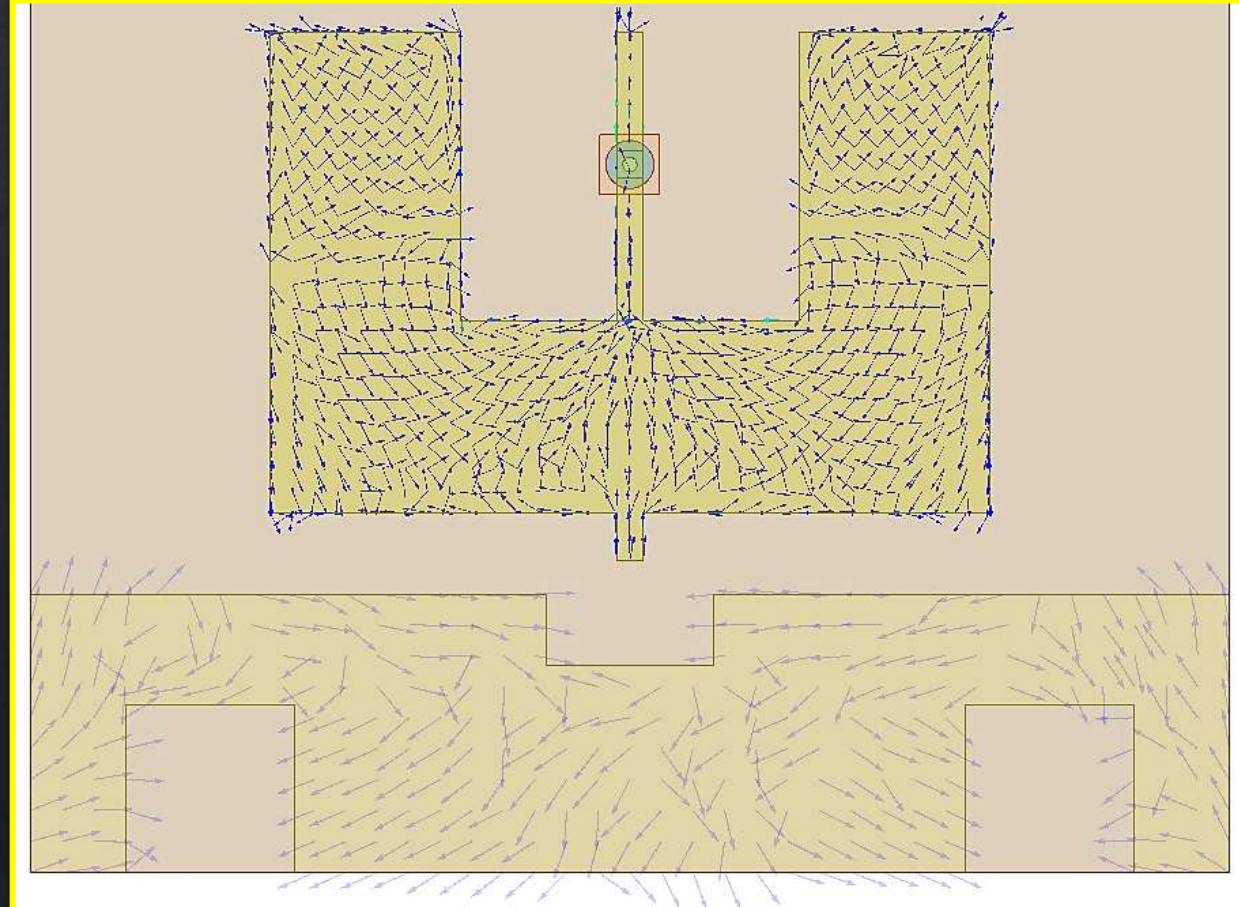
The S_{11} curve shows that the antenna achieves good impedance matching over a relatively wide frequency range. The **main resonant frequency** occurs at approximately **6.4 GHz**, where the reflection coefficient reaches a minimum of about **-31 dB**, indicating excellent matching at resonance. The antenna maintains S_{11} **below -10 dB** roughly from **6.0 GHz to about 8.8 GHz**, giving an impedance bandwidth of approximately **2.8 GHz**. The corresponding **fractional (percentage) bandwidth** is therefore around **35%**, calculated with respect to the center frequency. A secondary, weaker resonance can also be observed near **8.3 GHz**, which helps extend the bandwidth. Beyond this range, the matching gradually degrades, with S_{11} rising above -10 dB, indicating reduced efficiency outside the operating band.



Current distribution on the patch



Simulated surface current at first resonance (6.64 GHz)



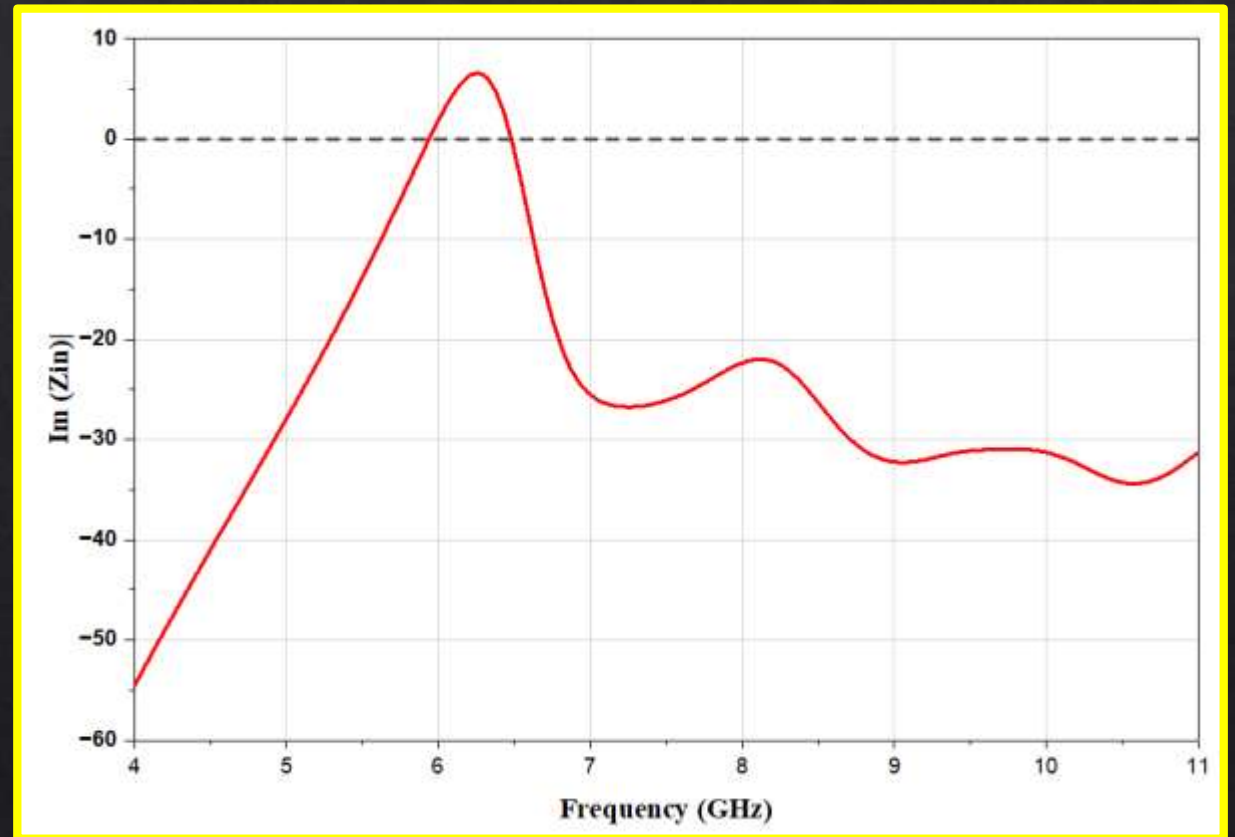
Simulated surface current at second resonance (8.4 GHz)

Simulation results

Input impedance vs frequency (Z_{in})

The imaginary part of the input impedance $\text{Im}(Z_{in})$ curve provides insight into the reactive behaviour of the antenna across frequency. At lower frequencies, the curve is negative, indicating a predominantly **capacitive nature**, and it gradually increases with frequency. The curve crosses the zero line at **5.89 GHz and 6.64GHz**, which corresponds to the **true resonant frequency** where the reactive component becomes zero and the antenna is purely resistive.

Beyond this point, the impedance briefly becomes inductive (positive), and then returns to a capacitive region at higher frequencies.



Simulation results

Broadside gain vs frequency

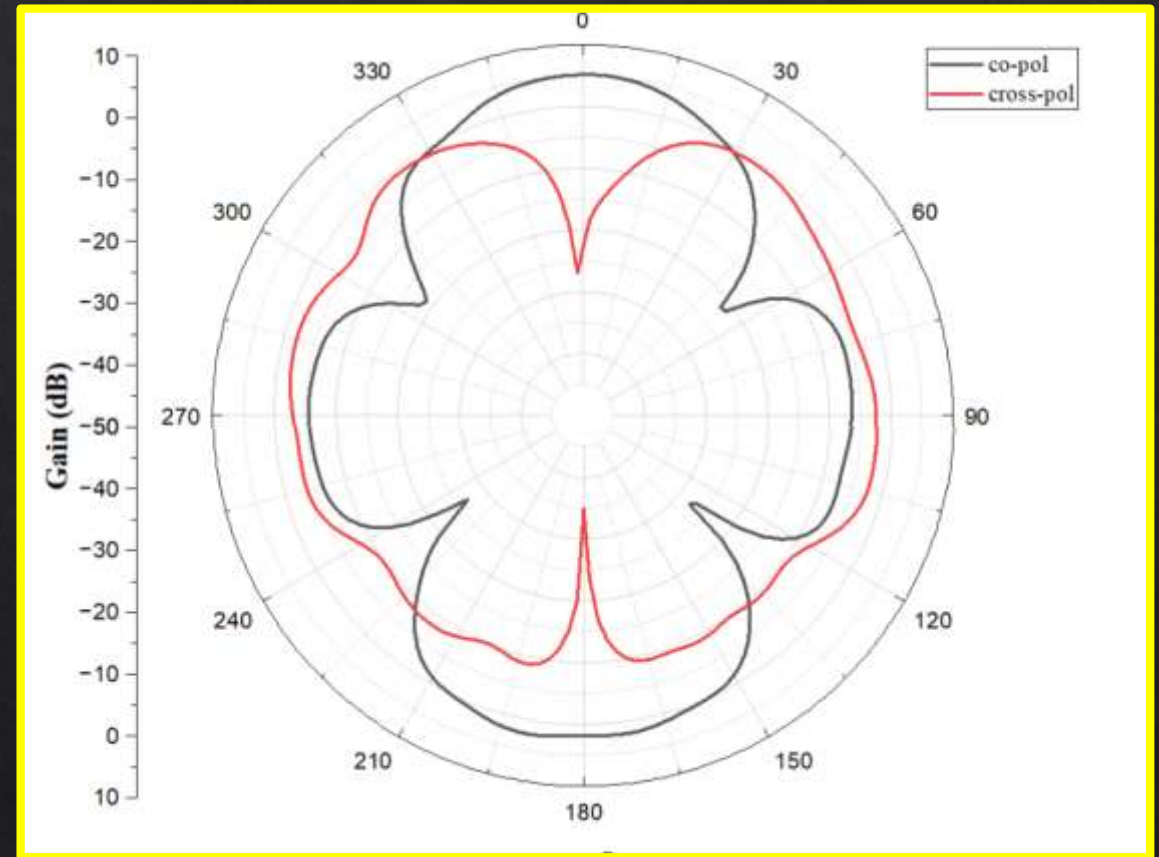
The broadside gain versus frequency curve shows that the antenna maintains a generally positive gain over most of the operating band, with values reaching around **4–5 dB** at higher frequencies, indicating efficient radiation in the broadside direction. However, a distinct dip in gain is observed around **6-7 GHz**, where the gain becomes negative. This negative gain corresponds to the presence of a **radiation null in the broadside direction** at that specific frequency, meaning that the antenna does not radiate effectively in that direction despite being excited. This behavior is typically caused by changes in current distribution or mode interaction at that frequency. An explanation of this phenomenon, along with the corresponding radiation characteristics, has been provided in the slide containing the **3D radiation pattern**, where the null formation can be clearly visualized.



Simulation results

Radiation pattern ($\phi=0$)

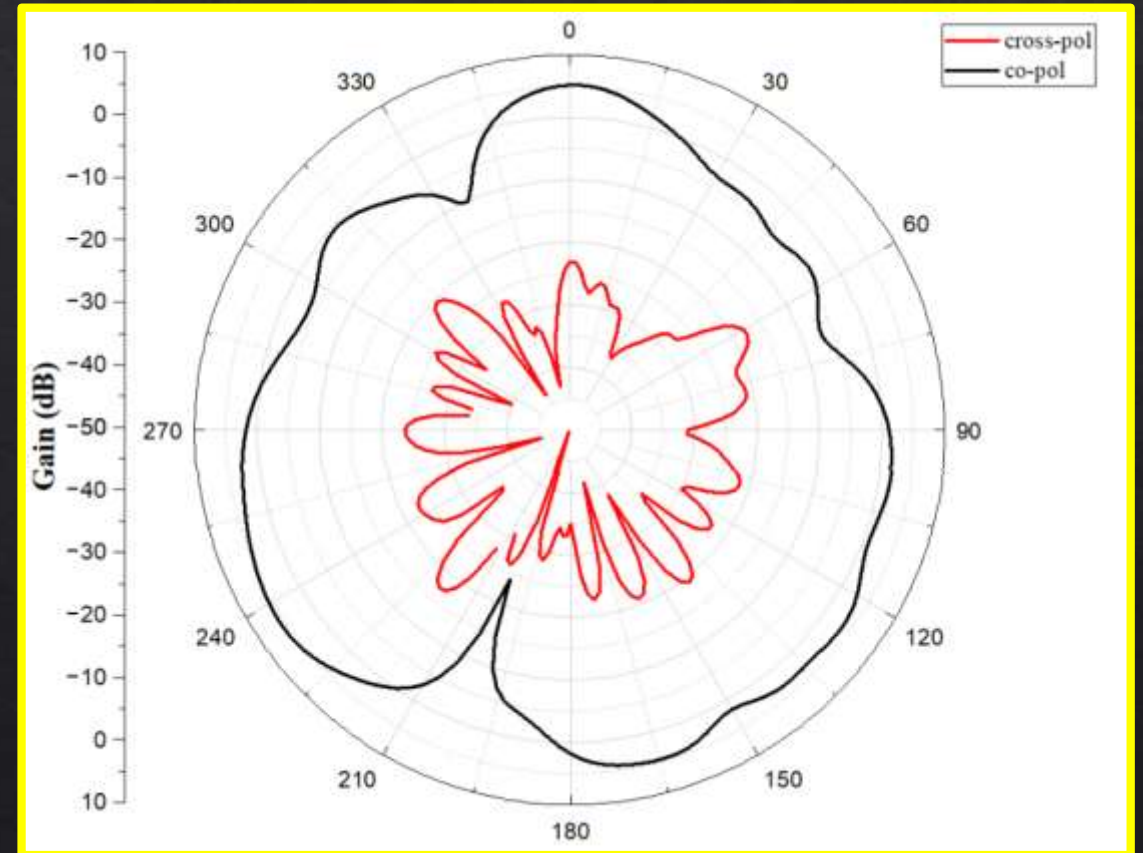
The radiation pattern of the antenna at $\phi = 0^\circ$ at the primary resonant frequency exhibits a **broadside radiation characteristic** with the main lobe directed normal to the plane of the antenna. The **co-polarized component** dominates across most angles, indicating that the antenna maintains the desired polarization and radiates efficiently in the principal direction. The pattern shows a reasonably wide beamwidth with smooth variation, suggesting stable radiation over a range of angles. The **cross-polarized component** remains significantly lower than the co-polarized field in the main lobe region, confirming good polarization purity. However, some minor lobes and variations are observed at larger angles, which may be attributed to higher-order mode contributions or structural asymmetries.



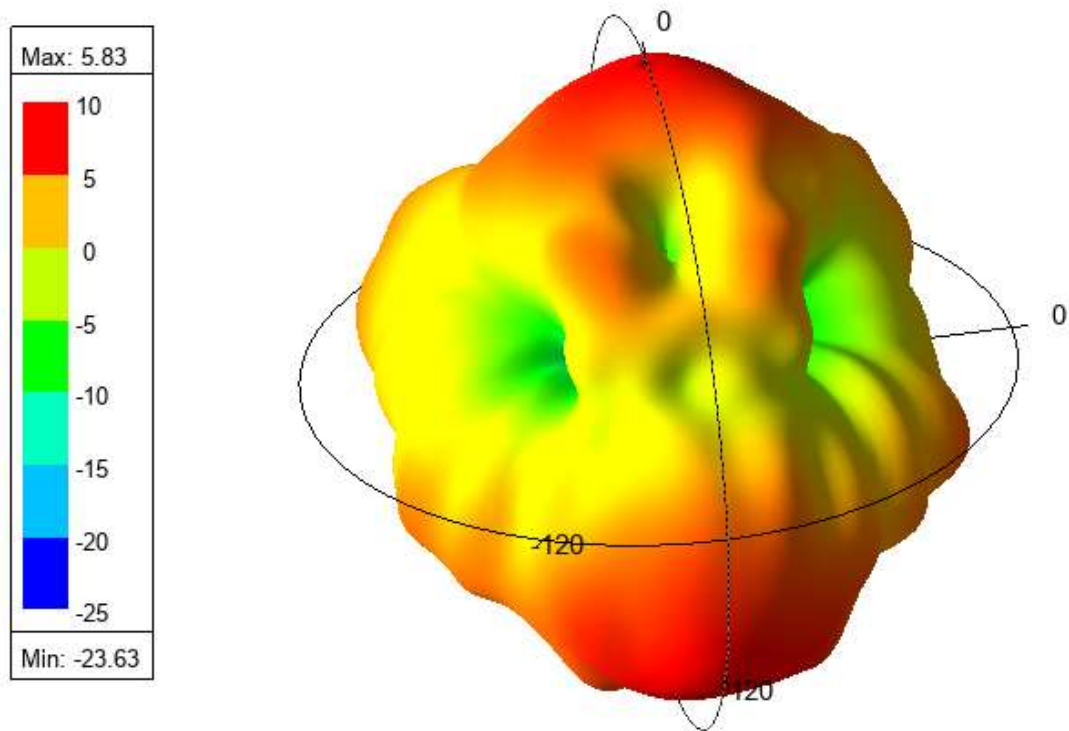
Simulation results

Radiation pattern ($\phi=90^\circ$)

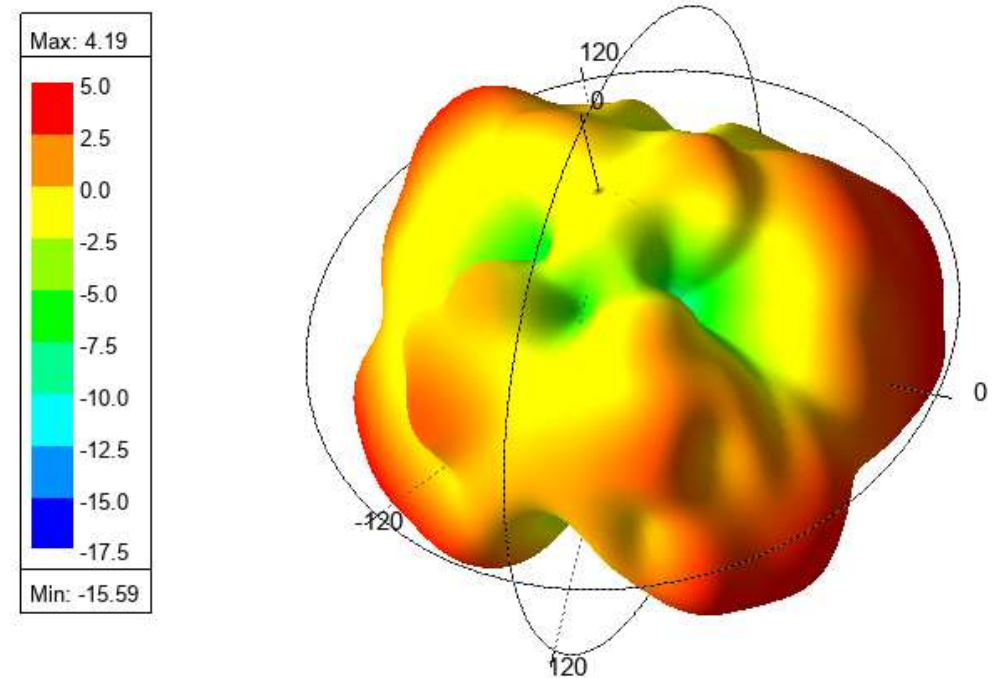
The radiation pattern of the antenna at $\phi = 90^\circ$ at the primary resonant frequency shows a **quasi-omnidirectional behavior** in the azimuth plane, which is characteristic of monopole-type radiation. The **co-polarized component** remains dominant over most angles, forming a nearly uniform pattern with slight variations and minor lobes, indicating reasonably consistent radiation in different directions. The **cross-polarized component** is significantly lower than the co-polarized field across the entire pattern, demonstrating good polarization purity. Some fluctuations and asymmetries in the pattern can be observed, which may arise due to structural features of the antenna or higher-order mode effects. Overall, the antenna exhibits stable radiation with predominantly omnidirectional characteristics in this plane at resonance.



Simulation results

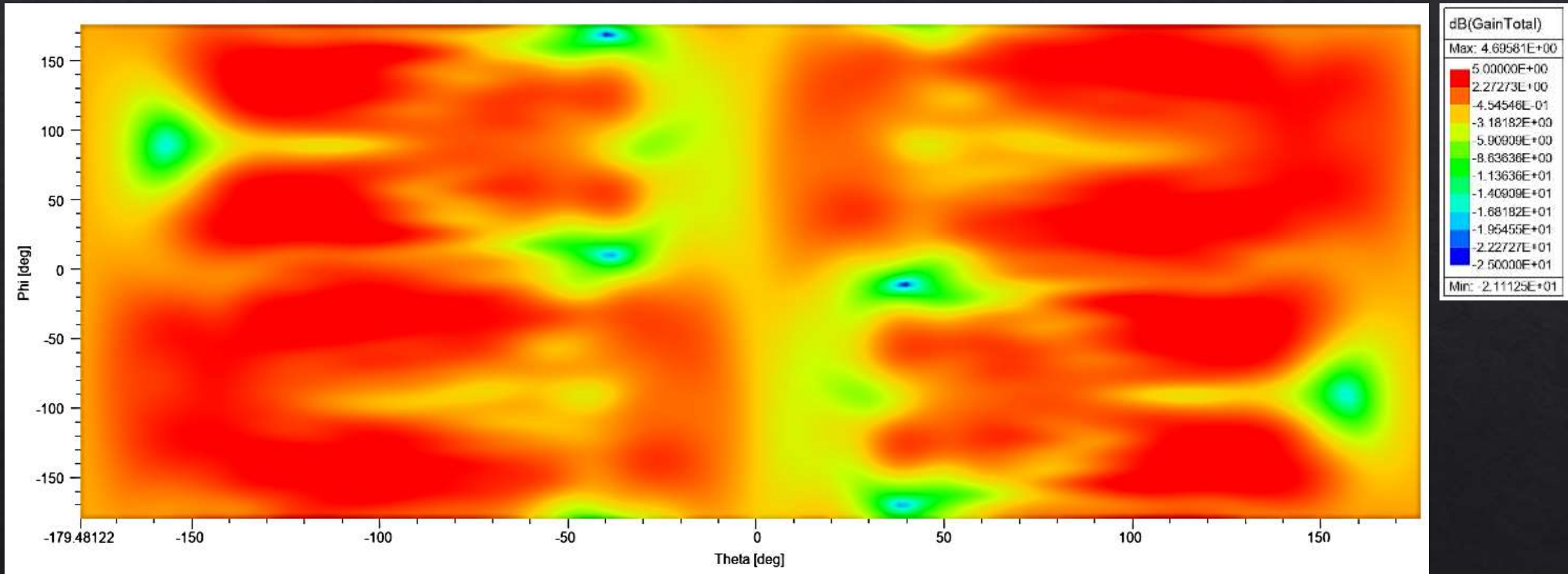


3D radiation pattern at first resonance (6.64 GHz)



3D radiation pattern at second resonance (8.4 GHz)

Simulation results



Gain contour plot against theta and phi showing the distribution of nulls at 6.64GHz

Conclusion

The designed printed monopole antenna demonstrates overall satisfactory performance, as evident from the impedance, gain, and radiation characteristics. The S_{11} results indicate good impedance matching over a wide bandwidth, supported by the presence of multiple resonances, while the gain plot shows stable radiation with only a localized dip corresponding to a radiation null. The radiation patterns in both $\phi = 0^\circ$ and $\phi = 90^\circ$ planes confirm the expected behavior of a monopole antenna, with dominant co-polarized radiation and acceptable cross-polarization levels. As expected for a printed monopole, radiation occurs on **both sides of the patch (front and back of the substrate)** due to the absence of a full ground plane, leading to a quasi-omnidirectional pattern. This bidirectional radiation behavior is consistent with the observed patterns, where energy is distributed in multiple directions rather than being confined to a single side. Overall, the results validate that the antenna operates as a monopole, with proper current distribution, wideband matching, and radiation characteristics aligned with theoretical expectations.

Thank you